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E-University System Network Design

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DECLARATION

We hereby declare that our thesis is entirely our work and is true/original. We understand that if any plagiarism is detected at any stage, our group will be given a grade of F (Fail) and this may result in our Above Average Diploma grade being withdrawn.

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TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

1.0 Introduction

In today's digital age, universities rely heavily on robust and efficient network infrastructures to support a wide range of academic, administrative, and research activities. A well-designed network ensures seamless connectivity, enhances communication, and provides the necessary resources for students, faculty, and staff to achieve their educational and operational goals. This project focuses on designing a comprehensive university network using Cisco Packet Tracer, a powerful network simulation tool, to meet the diverse needs of the university community. The proposed network design will address the key requirements of reliability, scalability, security, and cost-effectiveness, ensuring that the network can support current demands and future growth.

1.1 University Network Needs

universities have unique network requirements driven by their diverse and dynamic environments. These needs include:

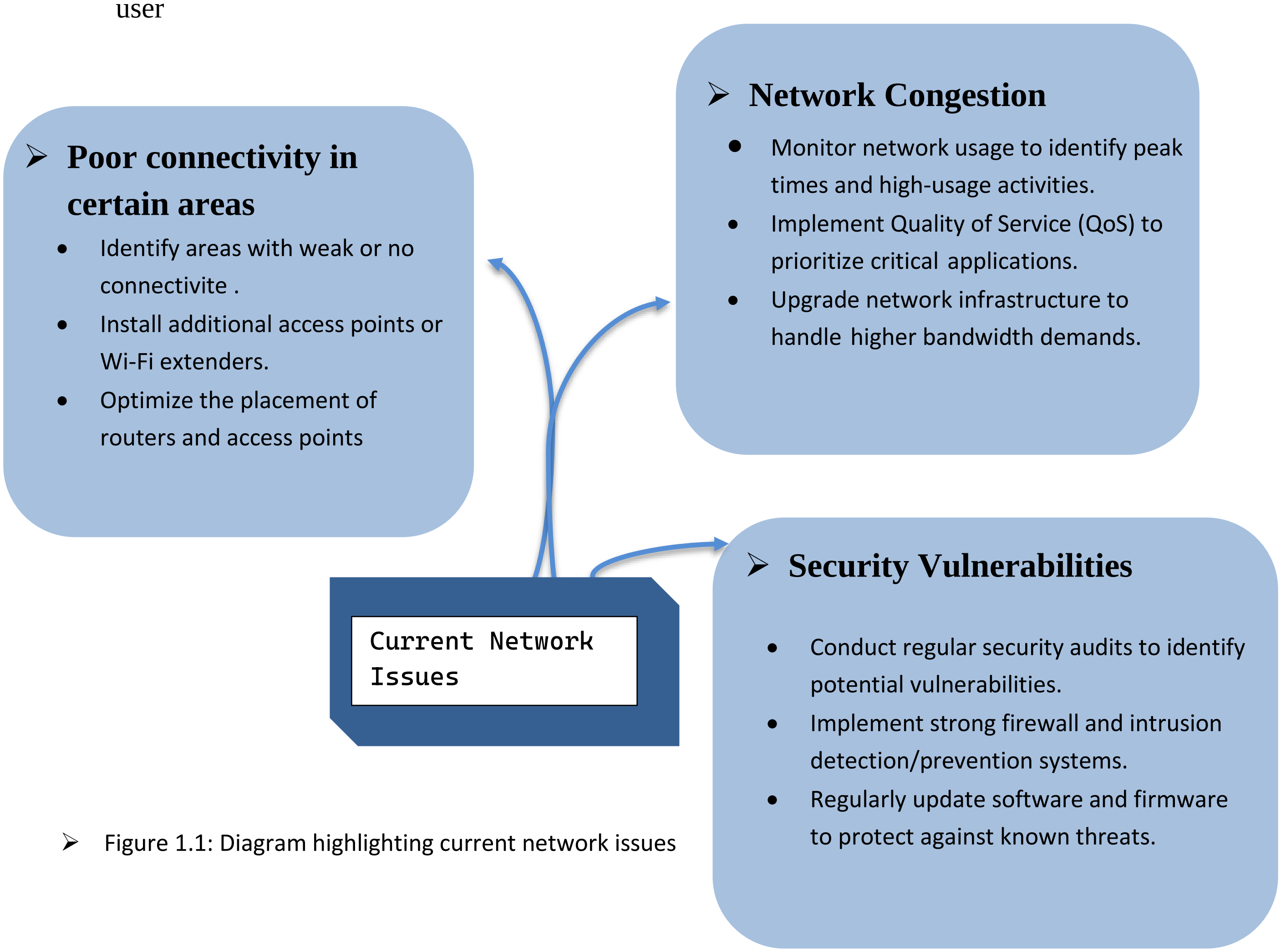
- **Reliability:** Ensuring uninterrupted network access for critical applications such as online learning platforms, administrative systems, and research databases.
- **Scalability:** Supporting the growing number of devices and users, as well as the increasing demand for high-bandwidth applications
- **Security:** Protecting sensitive data, such as student records and research information, from cyber threats through robust security measures.
- **Cost-Effectiveness:** Implementing a network design that maximizes performance and reliability while staying within budget
- **Constraints Flexibility:** Accommodating a mix of wired and wireless connectivity options to provide seamless access across the campus.
- **Support for Advanced Applications:** Enabling advanced applications such as VoIP, video conferencing, and collaborative research tools.

- **Ease of Management:** Utilizing network management tools and protocols to simplify administration, monitoring, and troubleshooting

Addressing these needs is critical for maintaining an efficient, secure, and scalable network that supports the university's mission of education and research

1.2 Problem Statement

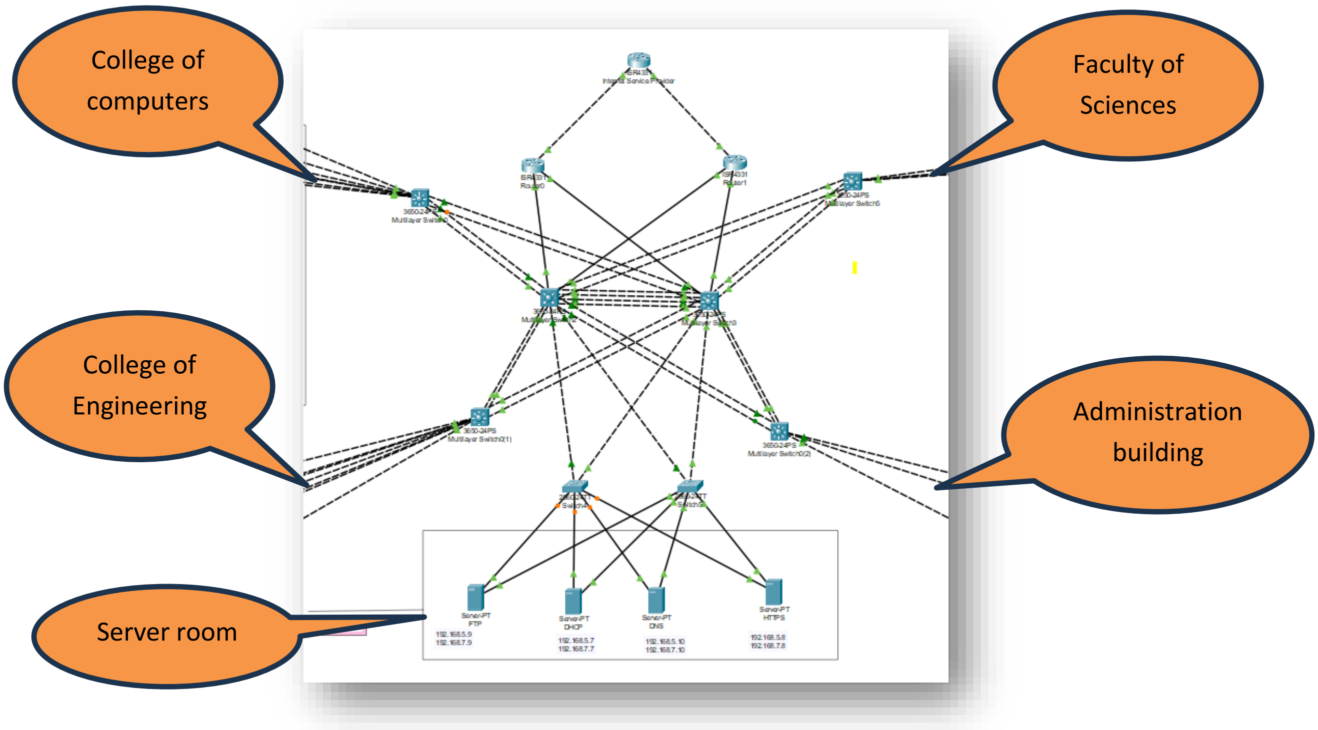
The current university network suffers from several issues, including poor connectivity in certain areas, network congestion, and security vulnerabilities. These issues hinder the effective functioning of academic and administrative processes, impacting the overall performance and user



➢ Figure 1.1: Diagram highlighting current network issues

1.3 Proposed Network Design Using Packet Tracer

To address these issues, we propose a new network design using Cisco Packet Tracer. This design will incorporate modern networking principles and technologies to ensure a robust, scalable, and secure network infrastructure



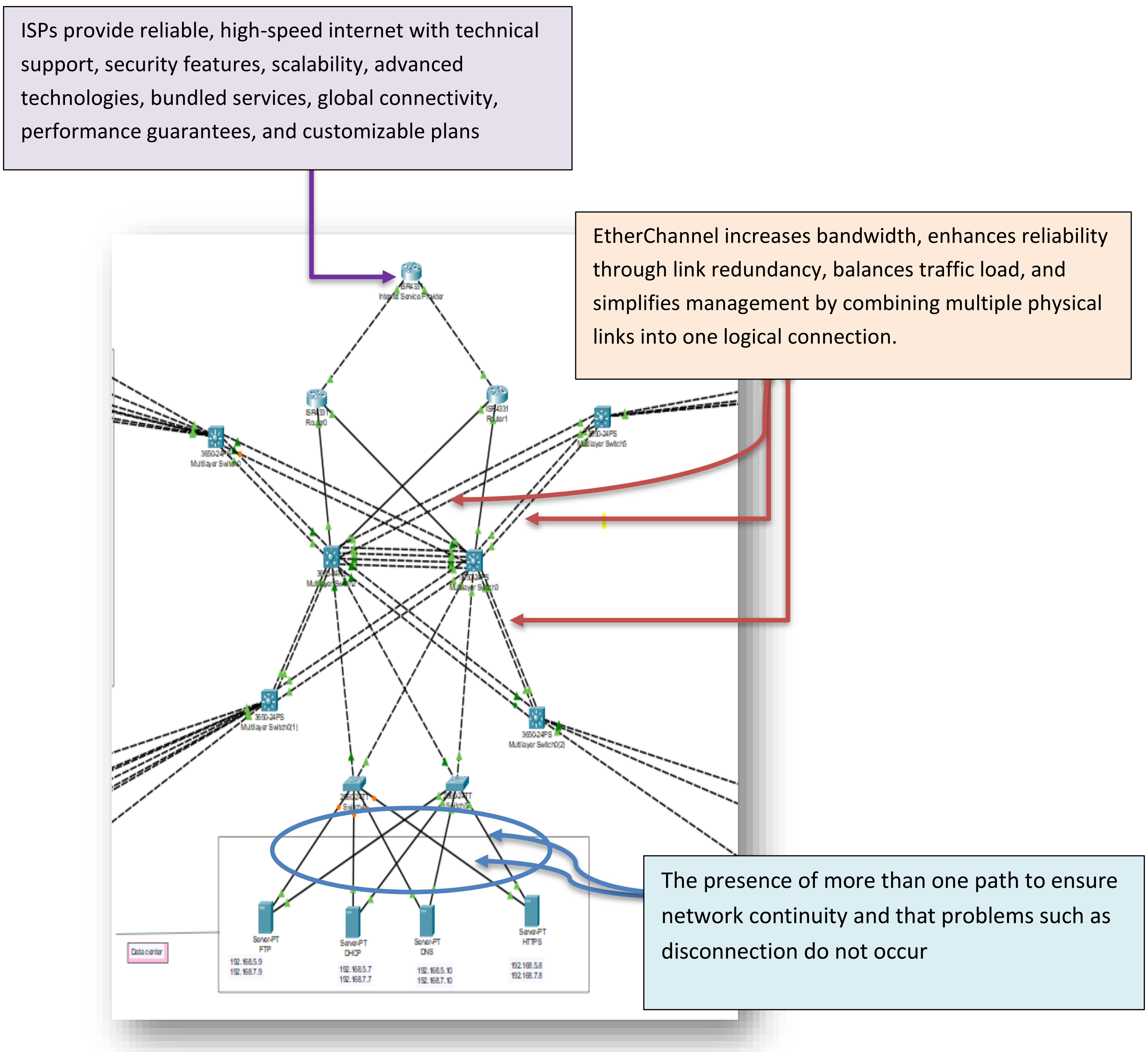
➢ Figure 1.2: Overview of the proposed network architecture

1.3.1 Aims and Objectives

- Ensure high reliability and uptime.
- Provide scalability to accommodate future growth.
- Implement robust security measures.
- Optimize cost-effectiveness without compromising quality.

1.3.2 Proposed System Features

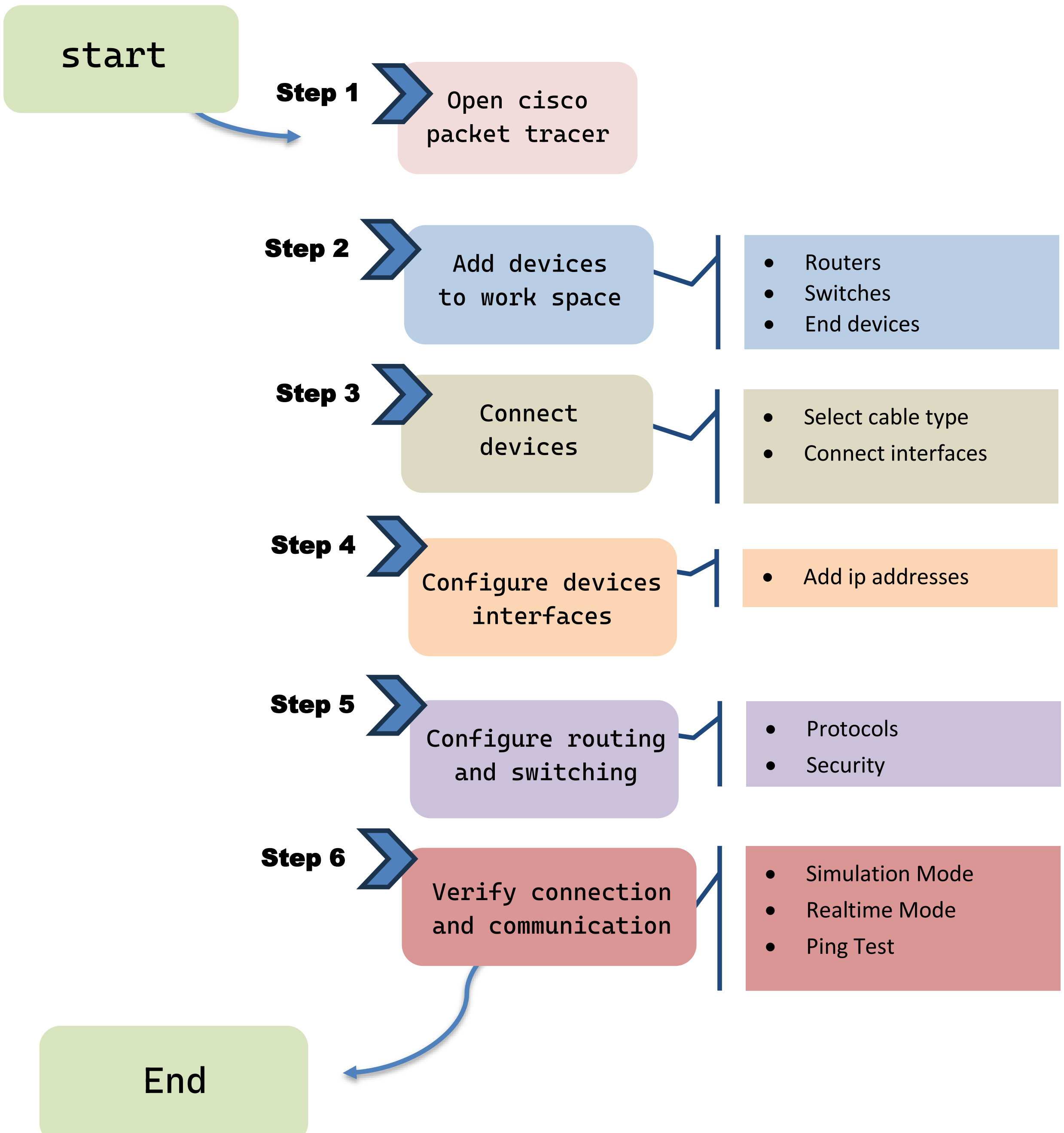
- Redundancy: Dual routers and switches to ensure failover.
- Security: Multi-layer security with firewalls, IDS/IPS, and VLAN segmentation.
- Performance: High-speed backbone to support data-intensive applications.



➢ Figure 1.3: Diagram showing key features of the proposed network

1.4 Project Methodology

The project will follow a structured methodology, including steps such as requirements gathering, design, implementation, and testing. This approach ensures a systematic and thorough development process.



- Figure 1.4: Flowchart illustrating the project methodology steps.

1.5 Resource Requirements

Hardware:
Routers, switches, multilayer switches, access points, servers, ip phone and PCs

Software:

Cisco Packet Tracer, network management tools.

Personnel:

Network engineers, IT staff, project manager.

1.6 Report Layout

This report is structured into several chapters, each focusing on a specific aspect of the network design. The subsequent chapters will delve into the background, design requirements, wireless network design, network services, management, case studies, future trends, and the conclusion.

The report is structured as follows:

Chapter 1: Introduction

Chapter 2: Background/Existing Work

Chapter 3: Network Design Requirements

Chapter 4: Wireless Network Design Using Packet Tracer

Chapter 5: Network Services and Applications

Chapter 6: Network Management and Monitoring

Chapter 7: Conclusion & Future Work

Network Diagram

Appendices

Glossary of Terms

Sources and Further Reading Recommendations

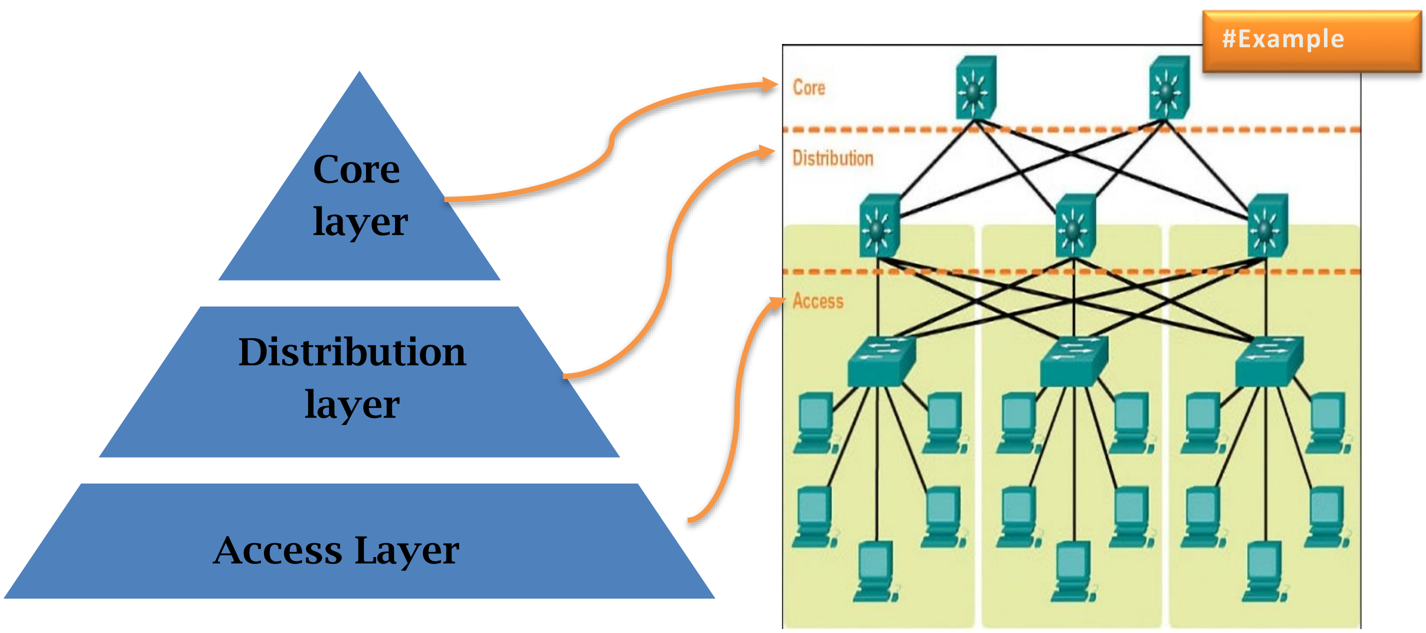
2.0 Introduction

This chapter explores the current landscape of university network infrastructures, identifying common practices and highlighting both strengths and limitations. Understanding existing setups is crucial for developing an improved network design that addresses specific challenges and leverages modern technologies.

2.1 Overview of Existing University Networks

University networks generally follow a hierarchical structure to ensure efficient data flow and manageability. Common components of these networks include:

- **Core Layer**: High-capacity switches and routers forming the backbone of the network, providing fast, reliable connectivity between different parts of the campus.
- **Distribution Layer**: Intermediate layer that aggregates data from the access layer, handles routing, filtering, and WAN access.
- **Access Layer**: Connects end devices like computers, printers, and wireless access points to the network, facilitating user access to network resources.



➢ Figure 2.1: Typical hierarchical structure of university networks.

2.2 Limitations of Existing Networks

- **Scalability Issues**: Many networks struggle to handle the increasing number of devices and growing data traffic, leading to congestion and reduced performance.

- **Security Vulnerabilities**: Outdated security protocols leave networks vulnerable to cyber threats, risking sensitive information such as student records and research data.

- **Resource Management Inefficiencies:** Poor management of IP addresses and network resources can result in conflicts and inefficiencies, impacting overall network performance.

- **Inadequate Wireless Coverage**: Insufficient Wi-Fi coverage in certain areas affects connectivity, frustrating users and limiting access to network resources.

➢ Figure 2.2 : Diagram highlighting common issues in existing university networks.

2.3 Innovations in Our Network Design

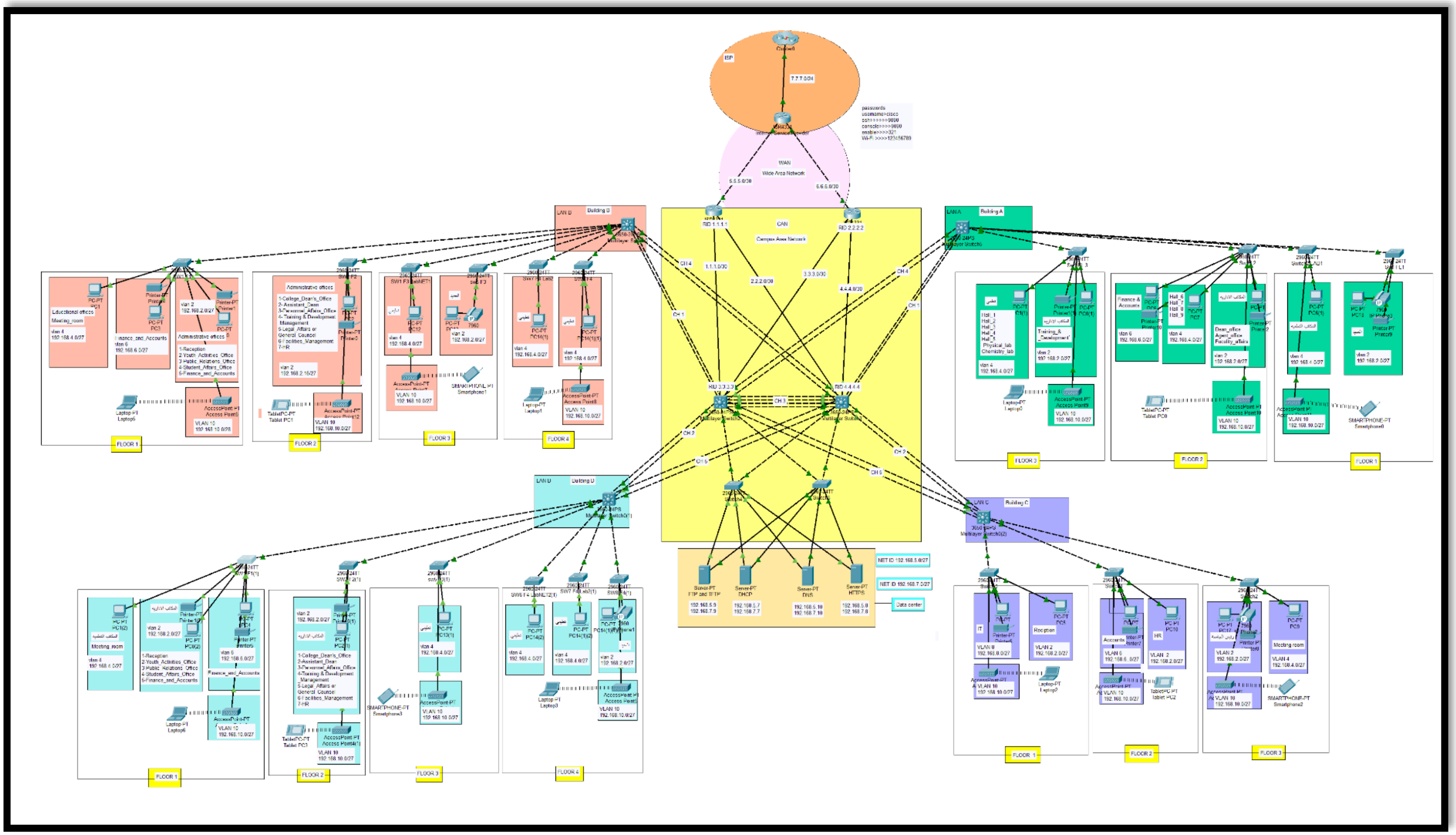
Our proposed network design aims to overcome these limitations by incorporating several key innovations:

- **Enhanced Scalability**: Implementing a modular design allows for easy expansion and upgrades, ensuring the network can grow alongside the university's needs.

- **Improved Security:** Deploying advanced security measures, including multi-layer security protocols with firewalls, IDS/IPS, and robust authentication mechanisms, to protect against cyber threats.

- **Efficient Resource Management**: Utilizing advanced IP address management tools and automated network management systems to optimize resource allocation and reduce conflicts.

- **Comprehensive Wireless Coverage**: Ensuring consistent and reliable Wi-Fi coverage across the entire campus using the latest wireless technologies (e.g., 802.11ac, 802.11ax) and effective RF management strategies.



➢ Figure 2.3: Overview of the proposed innovative network design.

3.0 Introduction

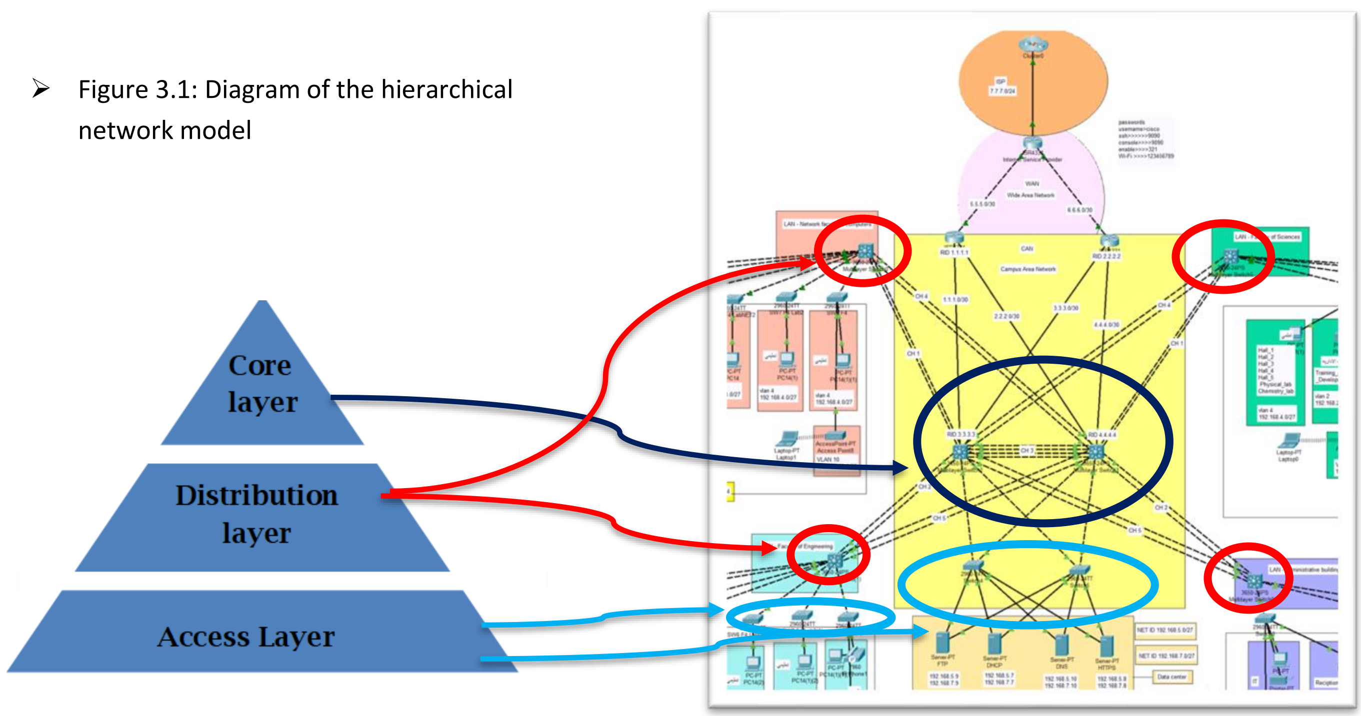
This chapter outlines the specific requirements for designing a robust and efficient university network. It covers the hierarchical network model, infrastructure needs, technologies and protocols, addressing and naming conventions, and security considerations.

3.1 Hierarchical Network Model

The network design will follow a hierarchical model to ensure scalability, manageability, and performance. This model includes three layers:

- **Core Layer:** High-speed backbone connecting different parts of the network.
- **Distribution Layer:** Aggregates data from the access layer and provides routing, filtering, and WAN access.
- **Access Layer:** Provides end-user connectivity through wired and wireless connections.

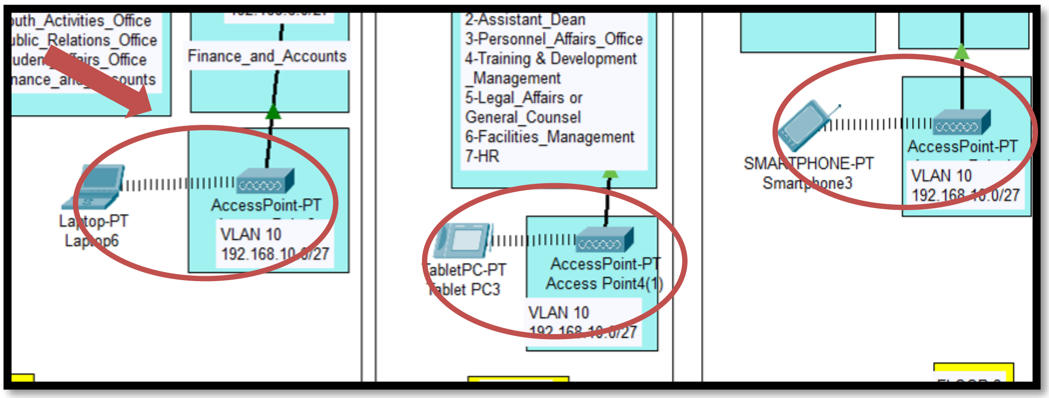
➢ Figure 3.1: Diagram of the hierarchical network model



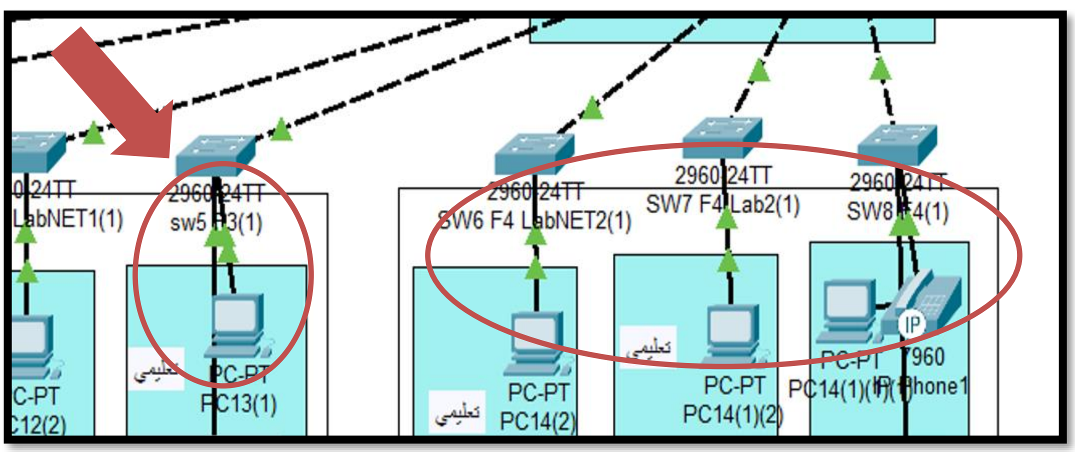
3.2 Infrastructure Requirements

3.2.1 Wired and Wireless Network Balance

A balanced approach between wired and wireless networks ensures flexibility and optimal performance across the campus.



➢ Figure 3.2: Diagram showing some wireless parts

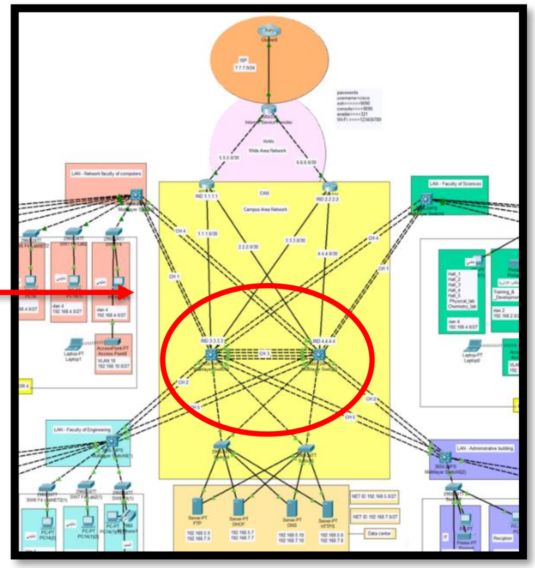


➢ Figure 3.3: Diagram showing some wired parts

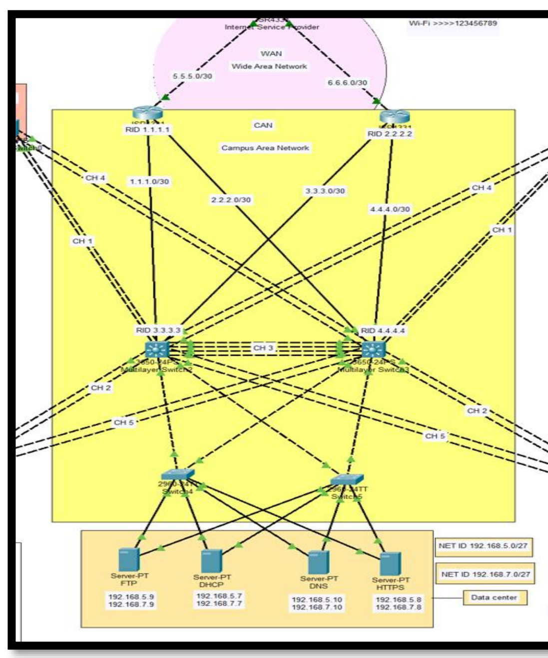
3.2.2 Robust Campus Backbone

A comprehensive and robust campus backbone, equipped with high-capacity switches and routers, is essential for ensuring reliable and high-speed connectivity across the entire campus network. This infrastructure supports the seamless flow of data, enabling efficient communication and access to resources for students, faculty, and staff. High-performance networking equipment reduces latency, minimizes downtime, and enhances the overall user experience, fostering an environment where academic and administrative activities can thrive without interruption.

BACKBONE OF NETWORK



- Figure 3.4: Diagram showing the backbone of the network



- Figure 3.5: Diagram showing the network infrastructure.

3.2.3 EtherChannel

EtherChannel is a port link aggregation technology used to combine multiple physical Ethernet links into a single logical link. This approach provides higher bandwidth and redundancy, enhancing both the performance and reliability of the network. By aggregating several physical connections, EtherChannel balances the load across the links, improving the network's efficiency and preventing a single point of failure.

Key Benefits of EtherChannel:

- **Increased Bandwidth:** Aggregates the capacity of multiple links, providing higher throughput than a single connection.
- **Load Balancing:** Distributes traffic across the bundled links, optimizing the use of available bandwidth.
- **Redundancy:** Enhances network reliability by ensuring that if one link in the EtherChannel bundle fails, the traffic is automatically rerouted to the remaining active links without disruption.
- **Scalability:** Simplifies network expansion as additional physical links can be added to the EtherChannel without reconfiguring the network topology.

EtherChannel Components:

1. **Physical Links:**

   Multiple Ethernet cables and ports that are bundled together to form a single logical connection.

2. **Logical Link:**

   The combined link that is treated as a single entity in terms of network configuration and management.

3. **Aggregation Protocols:**

   Port Aggregation Protocol (PAgP): Cisco proprietary protocol that automatically negotiates the formation of an EtherChannel.

   Link Aggregation Control Protocol (LACP): IEEE standard (802.3ad) protocol that provides similar functionality and is compatible with non-Cisco devices.

Types of EtherChannel:

There are three main types of EtherChannel, each with specific characteristics and use cases:

1. **Static (Manual) EtherChannel:**

   Description: Links are manually configured to form a channel without using any negotiation protocols.
   Use Case: Suitable for simple networks where dynamic negotiation is not necessary, requiring consistent configuration on both ends of the channel.

2. **Port Aggregation Protocol (PAgP):**

   Description: A Cisco proprietary protocol that automatically negotiates the formation of an EtherChannel. PAgP ensures that both ends of the channel are correctly configured and compatible.
   Use Case: Best for networks predominantly using Cisco devices, simplifying configuration and maintenance with automatic negotiation.

3. **Link Aggregation Control Protocol (LACP):**

   Description: An IEEE 802.3ad standard protocol used for dynamically negotiating and managing EtherChannel links. LACP ensures compatibility and correct configuration between devices from different vendors.
   Use Case: Ideal for multi-vendor environments where interoperability is essential. LACP provides flexibility and ease of management in dynamic network topologies.

EtherChannel Modes:

Each type of EtherChannel can operate in different modes, which determine how the links are aggregated and negotiated

- **Static (Manual) EtherChannel:**
  On Mode: EtherChannel is formed unconditionally without any negotiation. Both ends must be set to “on” mode.
  Off Mode: Disables EtherChannel, leaving interfaces as standalone links without aggregation or negotiation

- **PAgP Modes:**
  Auto: The interface responds to PAgP packets but does not initiate them. It forms an EtherChannel if it receives PAgP packets from the other side.
  Desirable: The interface actively initiates PAgP negotiation to form an EtherChannel.

- **LACP Modes:**
  Passive: The interface responds to LACP packets but does not initiate them. It forms an EtherChannel if it receives LACP packets from the other side.
  Active: The interface actively initiates LACP negotiation to form an EtherChannel.

How EtherChannel with LACP Works:

Link Aggregation:

Multiple physical links between two devices (e.g., switches) are aggregated into a single logical link.

Negotiation:

Using LACP, the devices negotiate the formation of the EtherChannel. One or both ends must be set to active mode for negotiation to occur.

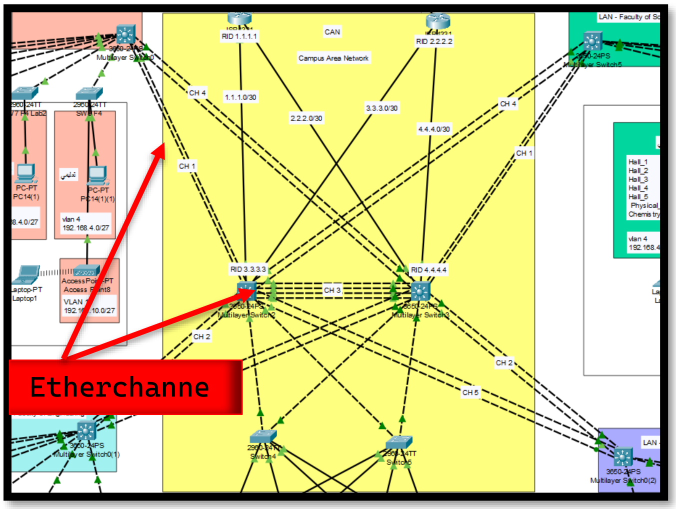
Traffic Distribution:

Traffic is distributed across the physical links using a hashing algorithm based on source and destination IP addresses, MAC addresses, or other criteria.

Failover:

If a physical link within the EtherChannel fails, LACP automatically redistributes traffic among the remaining links without interrupting the network connection.

EtherChannel with LACP is used to connect core switches to distribution switches, providing high bandwidth and redundancy. For instance, four Gigabit Ethernet links between the core switch and a distribution switch are bundled into a single 4 Gbps EtherChannel link. This setup not only increases the available bandwidth but also ensures that network connectivity is maintained even if one or more of the physical links fail.



➢ Figure 3.6: Diagram showing the EtherChannel

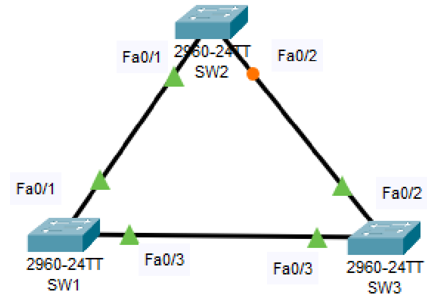
LACP ensures that the negotiation and maintenance of these links are handled automatically and efficiently.

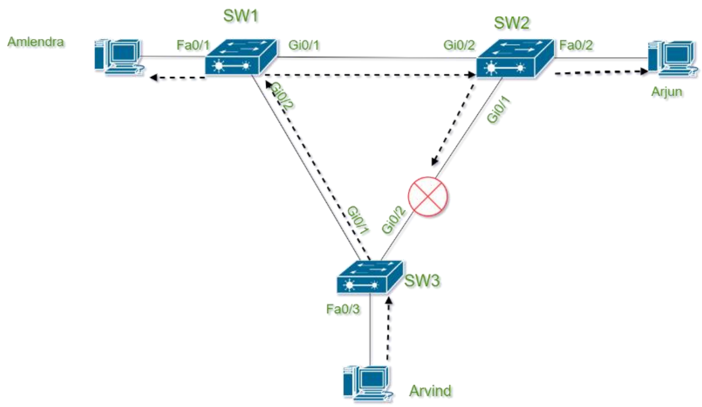
Implementing EtherChannel with LACP in our university's network infrastructure is crucial for enhancing bandwidth and providing redundancy.

By aggregating multiple physical links into a single logical connection, EtherChannel with LACP simplifies management, balances network load, and ensures continuous network availability. This standards-based approach ensures compatibility with a wide range of network devices, maintaining a robust and high-performance network environment for students, faculty, and staff.

3.2.4 Spanning Tree Protocol (STP)

Spanning Tree Protocol (STP) is a network protocol that ensures a loop-free topology for Ethernet networks. It is designed to prevent bridge loops and the broadcast radiation that results from them by creating a spanning tree within a network of connected layer-2 bridges (switches). STP operates by selectively blocking certain redundant paths in the network, thereby preventing data loops while ensuring network redundancy.





- Figure 3.7: STP Operation Diagram showing blocked links to prevent loops.

Types of Spanning Tree Protocols

There are several variations of STP, each offering different features and improvements over the original protocol:

1. Traditional STP (IEEE 802.1D):

Description: The original Spanning Tree Protocol standard. It operates by creating a single spanning tree for the entire network.

Use Case: Suitable for small and simple networks but has slower convergence times.

2. Rapid Spanning Tree Protocol (RSTP, IEEE 802.1w):

Description: An evolution of STP that provides faster convergence times by immediately transitioning ports to the forwarding state without waiting for timers to expire.

Use Case: Ideal for larger and more dynamic networks requiring quick recovery from topology changes.

3. Multiple Spanning Tree Protocol (MSTP, IEEE 802.1s):

Description: Allows multiple spanning trees to coexist in a single network, improving load balancing by mapping VLANs to different spanning tree instances.

Use Case: Best for complex networks with multiple VLANs needing optimized traffic distribution and redundancy.

4. Per-VLAN Spanning Tree Protocol (PVST and PVST+):

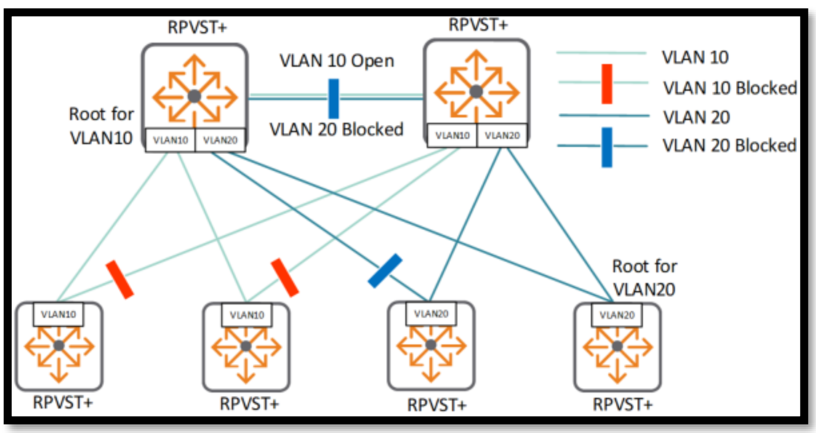
Description: Cisco proprietary protocols where each VLAN has its own spanning tree instance, providing better load balancing and more granular control.

Use Case: Suitable for networks with significant VLAN usage, enhancing performance and fault tolerance for each VLAN.

5. Rapid Per-VLAN Spanning Tree Protocol (RPVST+):

Description: A Cisco enhancement of RSTP that provides rapid convergence on a per-VLAN basis. Each VLAN runs its own RSTP instance.

Use Case: Ideal for large, VLAN-heavy networks requiring quick recovery and high performance across multiple VLANs.



- Figure 3.8: RPVST+ Operation Diagram demonstrating separate spanning tree VLAN

Implementation of RPVST+ in Our Network Design

In our university network design, we have chosen to implement Rapid Per-VLAN Spanning Tree Protocol (RPVST+). **This choice offers several key advantages:**

Rapid Convergence: RPVST+ significantly reduces the time required for the network to adapt to changes, ensuring minimal disruption during topology changes or failures.

Per-VLAN Optimization: Each VLAN operates its own spanning tree instance, allowing for optimized path selection and load balancing tailored to individual VLANs.

Enhanced Redundancy: The protocol provides efficient redundancy and failover mechanisms, maintaining network stability and availability.

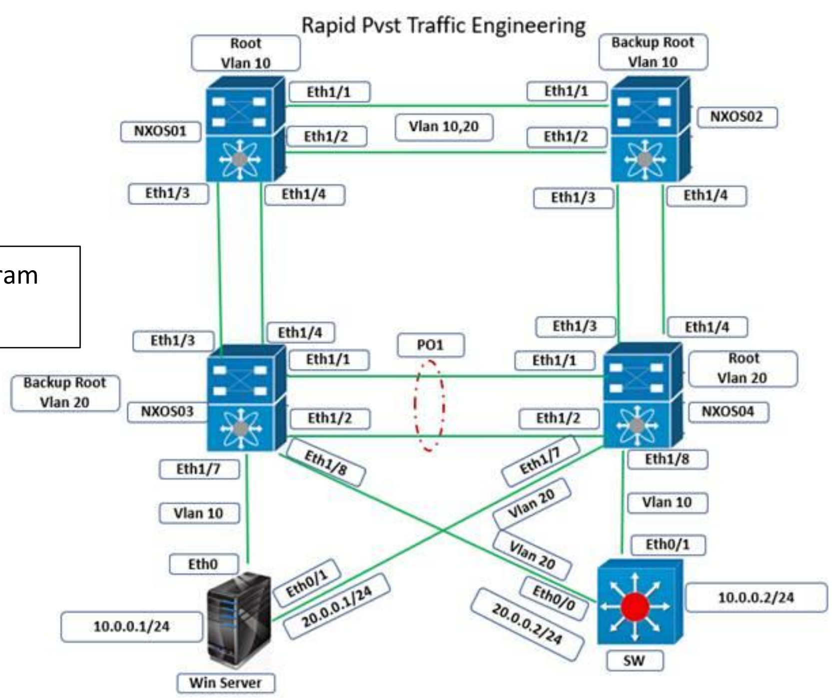
Key Benefits of RPVST+ Implementation

Improved Network Performance: By enabling rapid convergence and per-VLAN optimization, RPVST+ enhances overall network performance and efficiency.

Increased Reliability: The protocol ensures robust redundancy and quick recovery from failures, which is critical for maintaining continuous network services in a university environment.

Simplified Troubleshooting: With individual spanning trees for each VLAN, network administrators can more easily identify and resolve issues within specific segments of the network.

➢ Figure 3.9: RPVST+ Operation Diagram



In our university network, RPVST+ is used to manage the multiple VLANs implemented across different departments and buildings. By utilizing RPVST+, we ensure that each VLAN benefits from rapid convergence and optimized path selection, providing a stable and efficient network experience for students, faculty, and staff. For instance, if a link failure occurs in one VLAN, RPVST+ quickly recalculates the spanning tree for that VLAN without affecting other VLANs, ensuring minimal disruption and maintaining overall network performance.

The implementation of RPVST+ in our university's network infrastructure provides significant advantages in terms of rapid convergence, per-VLAN optimization, and enhanced redundancy. By leveraging this advanced spanning tree protocol, we ensure a robust, reliable, and high-performance network that meets the diverse needs of our academic and administrative users.

3.3 Network Technologies and Protocols

3.3.1 Ethernet and VLANs

To segment the network, improve performance, and enhance security, Ethernet and Virtual Local Area Networks (VLANs) will be strategically implemented across the network.

This VLAN configuration supports the university's goal of providing a reliable, secure, and efficient network environment for all users.

3.3.2 VLAN Configuration

To enhance network performance, security, and management, we have implemented several VLANs across the university campus. Each VLAN is designated for specific purposes to ensure efficient data traffic segregation and optimized resource allocation.

1. VLAN 10: Access Points:

- **Purpose:** Dedicated for wireless access points.
- **Benefits:** Ensures stable and efficient connectivity for all wireless devices.

2. VLAN 150: Voice:

- **Purpose**: Reserved for voice communication systems.
- **Benefits:** Provides high-quality, uninterrupted voice services by prioritizing voice traffic.

3. VLAN 4: Educational:

- **Purpose:** Designated for academic and educational resources.
- **Benefits:** Streamlines access to educational content and applications, enhancing the learning experience

4. VLAN 6: Accounts and Finance:

- **Purpose:** Allocated for the Accounts and Finance department.
- **Benefits:** Secures sensitive financial data and transactions, ensuring compliance and data integrity.

4. VLAN 2: Administrative offices:

- **Purpose:** Allocated to administrative offices for day-to-day operations.
- **Benefit:** Ensures efficient and secure access to administrative resources and data

By structuring the network with these VLANs, we achieve:

Improved Network Performance: By isolating different types of traffic, each VLAN can operate more efficiently.

Enhanced Security: Sensitive data, especially for the Accounts and Finance department, is protected within its own VLAN.

Better Management: Simplifies network administration and troubleshooting, allowing for quicker responses to issues.

3.3.3 Inter VLAN routing

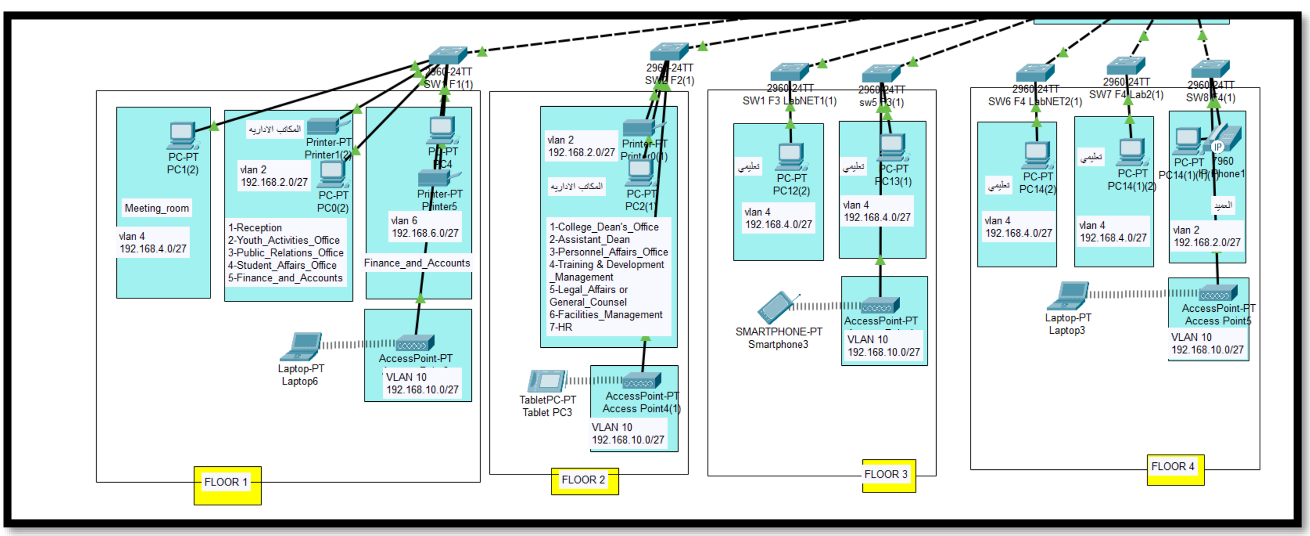
To enable efficient communication between different VLANs within our university network, we have implemented inter-VLAN routing using a Layer 3 switch. This setup allows devices on separate VLANs to communicate with each other seamlessly, ensuring optimal performance and security across the campus. Here is a detailed explanation of our inter-VLAN routing configuration:

Purpose of Inter-VLAN Routing

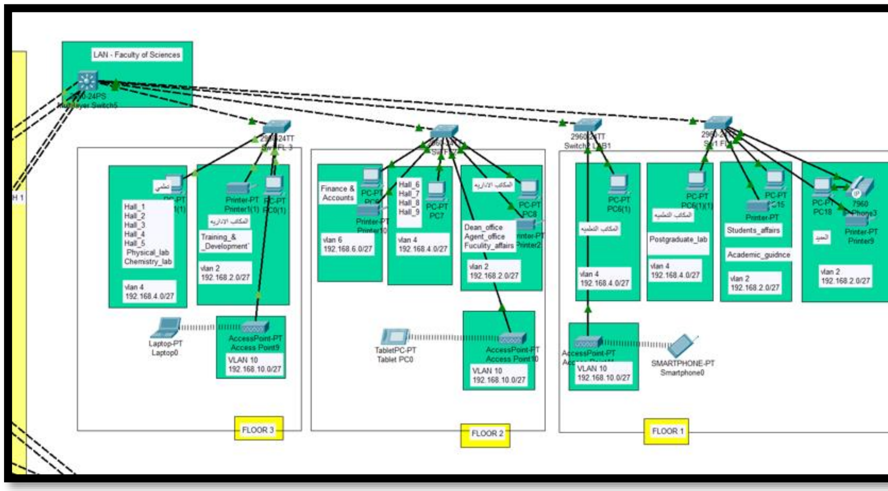
- **Enable Communication:** Allows devices on different VLANs to communicate, enhancing collaboration and resource sharing.
- **Improve Network Management:** Simplifies network management by logically segmenting traffic while maintaining connectivity.

Vlan name	Number of vlan	Purpose	subnet
Administrative offices	2	Allocated to administrative offices, ensuring efficient and secure access to administrative resources and data	192.168.2.0/27
Education	4	Used for academic purposes, including classrooms, labs, and student access, optimizing network resources for educational use.	192.168.4.0/27
Accounts and Finance	6	Reserved for the accounts and finance departments, enhancing security and performance for financial operations	192.168.6.0/27
Voice	150	Dedicated to voice communication systems, ensuring high-quality and reliable voice services	192.168.150.0/27
Access Points	10	Dedicated to wireless access points, providing robust and efficient wireless connectivity across the campus	192.168.10.0/27

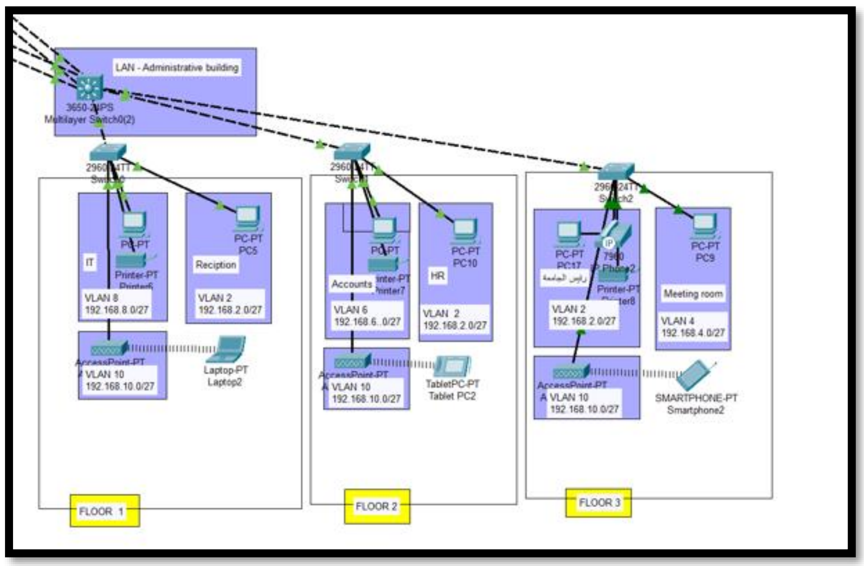
➢ Table 3.1: A table showing the configure VlAN in the network



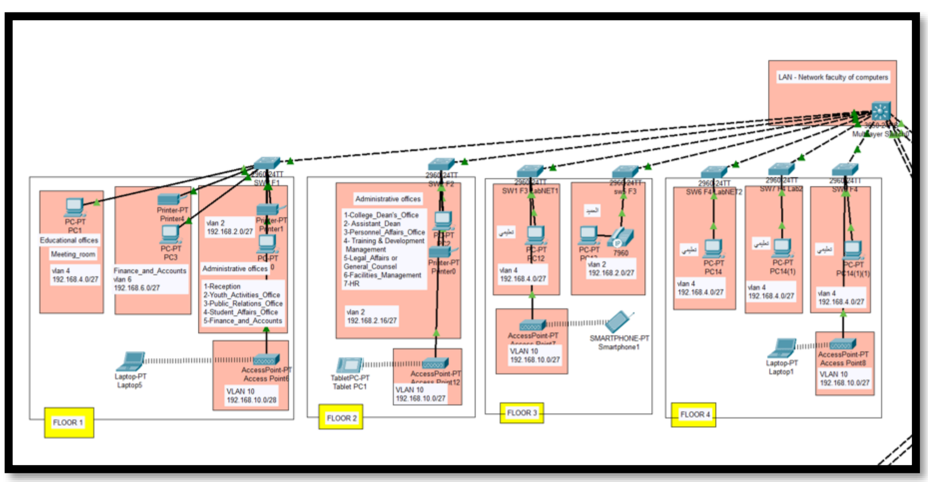
➢ Figure 3.10: Diagram showing the vlan configure of Faculty of Engineering



➢ Figure 3.11: Diagram showing the vlan configure of Faculty of sciences



➢ Figure 3.12: Diagram showing the vlan configure of administrative building



➢ Figure 3.13: Diagram showing the vlan configure of Network faculty of computers

3.3.4 Routing Protocols: OSPF

OSPF Configuration in Our University Network Using a Single Area (Area 0)

To ensure efficient and reliable routing within our university network, we have implemented OSPF (Open Shortest Path First) using a single area, Area 0. OSPF is ideal for our network due to its fast convergence, efficient use of network resources, and scalability.

Purpose of OSPF

- **Efficient Routing:** Determines the shortest and most efficient path for data to travel across the network.
- **Fast Convergence:** Quickly adapts to network changes, ensuring minimal disruption.
- **Scalability:** Supports large and complex network topologies, making it ideal for our growing university network

Benefits of OSPF

- **Dynamic Routing:** Automatically adjusts routes based on network changes, ensuring efficient data delivery.
- **Load Balancing**: Distributes traffic evenly across multiple paths, optimizing network performance.
- **Simplified Management:** Using a single area (Area 0) simplifies configuration and management while maintaining network efficiency

3.3.5 Network Address Translation (NAT)

Network Address Translation (NAT) is a technique used to modify network address information in the IP header of packets while they are in transit. NAT is essential for conserving global IP address space, enhancing security, and allowing multiple devices on a local network to share a single public IP address for accessing external networks

Key Benefits of NAT:

- **IP Address Conservation:** Allows multiple devices on a private network to share a single public IP address, reducing the need for numerous public IP addresses.
- **Security:** Hides internal IP addresses from external networks, protecting internal devices from potential threats and unauthorized access.
- **Simplified Network Management:** Enables easier network renumbering and management of IP address changes within a private network without affecting external communications.

Types of NAT:

Static NAT: Maps a single private IP address to a single public IP address. Used for devices that require a consistent public IP address, such as servers.
Dynamic NAT: Maps private IP addresses to a pool of public IP addresses, allowing internal devices to use any available public IP address from the pool for outbound communications.
Port Address Translation (PAT): Also known as Overloading, PAT maps multiple private IP addresses to a single public IP address or a few addresses using different ports. Commonly used in home and small office networks.

How NAT Works:

- **Translation Process**: When a device on the internal network sends a packet to an external network, the NAT device (usually a router) modifies the source IP address in the packet header to the public IP address assigned to the NAT device.
- **Maintaining Connections**: NAT maintains a translation table to keep track of active connections, ensuring that response packets from the external network are correctly translated back to the original internal IP address.
- **Outbound and Inbound Traffic:** While NAT is primarily used for outbound traffic (from internal to external networks), it can also handle inbound traffic by mapping specific external ports to internal devices (port forwarding).

In our university network, PAT is used to enable multiple devices within the campus to access the internet using a limited number of public IP addresses. This approach conserves IP address space and enhances network security by masking the internal IP addresses of devices.

Implementing NAT, specifically PAT, in our university's network infrastructure is crucial for efficient IP address management, enhanced security, and seamless connectivity between internal and external networks. PAT allows us to maximize the use of available IP addresses while protecting internal devices from external threats.

3.3.6 Remote Access

Remote access technologies are essential for allowing users to connect to the university network from remote locations. These technologies ensure that faculty, staff, and students can securely access the network resources they need, regardless of their physical location. Remote access is typically implemented using Virtual Private Network (VPN) technologies, which provide secure connections over public networks

key Benefits of Remote Access:

- **Flexibility:** Enables users to access network resources from anywhere, at any time, which is especially beneficial for remote learning and telecommuting.
- **Security:** Ensures data is encrypted and secure during transmission over public networks, protecting sensitive information from interception.
- **Productivity:** Allows users to work remotely without losing access to essential tools and resources, thereby maintaining productivity

Types of Remote Access:

Remote Desktop Access:

Description: Allows users to remotely control a desktop or workstation as if they were physically present at the machine.

Use Case: Faculty and staff can remotely access their office from home to work on projects and access software installed only on their office machines.

Virtual Private Network (VPN):

Description: Establishes a secure connection over the internet to the university's network, allowing users to access network resources as if they were on campus.

Use Case: Used by faculty, staff, and students to securely access internal network resources and services from remote locations.

Secure Shell (SSH):

Description: Provides a secure way to access and manage network devices and servers remotely via command-line interface.

Use Case: IT staff and researchers can remotely manage servers and network infrastructure securely, ensuring that administrative tasks can be performed from any location.

Web-Based Access:

Description: Allows users to access certain applications and services through a web browser, often using secure HTTPS connections.

Use Case: Access to web-based applications such as email, learning management systems, and administrative portals from any internet-connected device.

How Remote Access Works with SSH:

SSH Client Software: Users install SSH client software on their devices. This software allows them to initiate a secure connection to the university's SSH server.

Authentication: Users authenticate themselves using credentials (such as a username and password) or other methods like key-based authentication to ensure that only authorized users can access the network.

Secure Connection: Once authenticated, the SSH client establishes a secure, encrypted connection to the university network, ensuring that all data transmitted between the user's device and the network is protected.

SSH is used to enable faculty, staff, and students to securely access campus resources from home or other remote locations. This is particularly useful for accessing servers, administrative systems, and research databases, ensuring that users can continue their work without interruption.

Implementing various remote access technologies, including SSH, in our university's network infrastructure is crucial for providing secure, flexible, and reliable connectivity for users. By utilizing these technologies, we can ensure that faculty, staff, and students have secure access to the resources they need, regardless of their location, thereby supporting remote learning and telecommuting initiatives.

3.4 Addressing and Naming

Effective addressing and naming are crucial for the efficient operation of our university network. This section outlines the strategies and configurations we use for IP address management, as well as the implementation of DNS and DHCP services

3.4.1 IP Address Management (IPv4)

Proper IP address management ensures that every device on our network has a unique and appropriately assigned IP address, facilitating smooth communication and minimizing conflicts.

IP Address Scheme:

- **Subnetting:** We have divided our network into smaller subnets using a /27 subnet mask to efficiently use IP address space and segment traffic.
- **VLANs and Subnets**: Each VLAN is associated with a specific IP subnet to logically separate different types of traffic and departments.

IP Address Allocation:

- **Static IP Addresses:** Reserved for critical network devices such as servers, routers, and switches.
- **Dynamic IP Addresses:** Assigned to end-user devices such as computers and smartphones devices via DHCP.

Example of IP Address Allocation:

Vlan	Subnet	Description
VLAN 2	192.168.2.0/27	Administrative Offices
VLAN 4	192.168.4.0/27	Educational
VLAN 6	192.168.6.0/27	Accounts and Finance
VLAN 10	192.168.10.0/27	Access Points
VLAN 150	192.168.150.0/27	Voice

➢ Table 3.2: A table showing the Example of IP Address Allocation

Subnet Mask: Each subnet uses a subnet mask of 255.255.255.224 (also represented as /27)

3.4.2 DNS and DHCP Implementation

To manage IP addresses and resolve hostnames within our network, we have implemented DNS and DHCP services.

❖ DNS (Domain Name System):

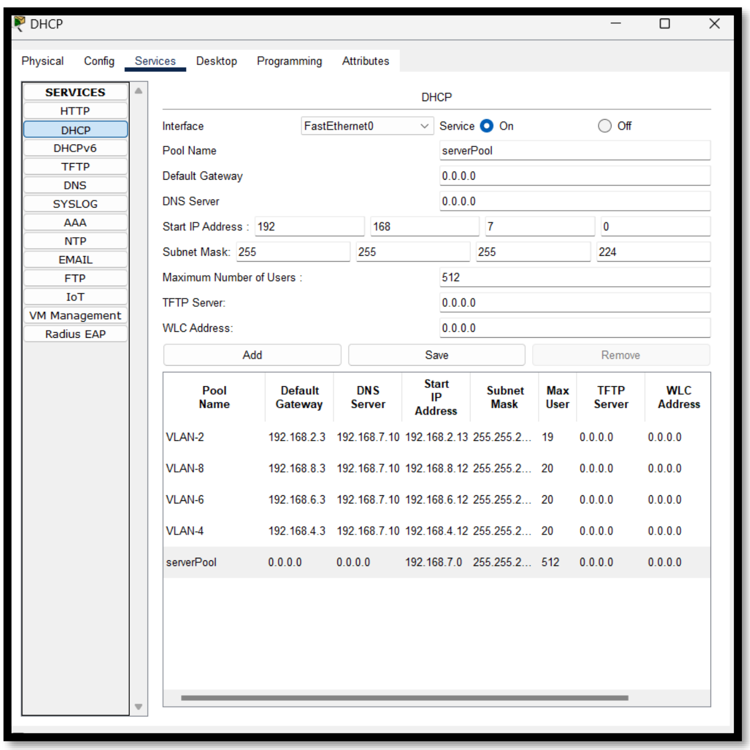
DNS translates human-readable domain names into IP addresses, allowing users to access network resources using easy-to-remember names.

❖ DHCP (Dynamic Host Configuration Protocol):

DHCP automates the assignment of IP addresses to devices on the network, reducing administrative overhead and preventing IP conflicts.

- **DHCP Server:** Configured to assign IP addresses, subnet masks, default gateways, and DNS server addresses to DHCP clients.

- **IP Address Pool:** Defined for each VLAN, ensuring devices receive appropriate IP addresses based on their VLAN.



➢ Figure 3.14: Diagram showing the DHCP server pools and configure

DNS Implementation:

Internal DNS servers resolve hostnames to IP addresses, simplifying access to network resources.

DHCP Implementation:

Automates IP address assignment, reducing conflicts and administrative workload.

By implementing robust IP address management, DNS, and DHCP services, we ensure a well-organized and efficient network infrastructure that supports the operational needs of our university.

3.5 Security Considerations

Ensuring the security of our university network is of paramount importance. This section outlines the measures and protocols we implement to safeguard the network from unauthorized access, misuse, and potential threats.

3.5.1 ACL - Port Security - DHCP Snooping

1. **Access Control Lists (ACLs):**

ACLs are used to control the flow of traffic into and out of the network. They help in permitting or denying traffic based on specific criteria such as IP addresses, protocols, and port numbers.

Control traffic flow and enhance security by filtering based on IP addresses and ports.

- **Standard ACLs**: Filter traffic based only on the source IP address.

- **Extended ACLs:** Provide more granular control by filtering traffic based on both source and destination IP addresses, as well as protocols and port numbers.

2. **Port Security:**

Port security is configured on switches to prevent unauthorized devices from connecting to the network. This feature limits the number of MAC addresses that can be learned on a single port.

Prevents unauthorized devices from connecting to the network by limiting the number of MAC addresses on switch ports.

- **Static Secure MAC Addresses:** Manually configured and allowed on the port.
- **Dynamic Secure MAC Addresses**: Learned dynamically but are lost when the switch reboots.
- **Sticky Secure MAC Addresses**: Learned dynamically and saved in the running configuration.

Example of Port Security Configuration:

plaintext Copy code

```
interface FastEthernet0/1
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict
```

3. DHCP Snooping:

DHCP snooping is a security feature that filters untrusted DHCP messages and builds a DHCP binding table that maps IP addresses to MAC addresses and switch ports. It helps in preventing rogue DHCP servers.

Protects against rogue DHCP servers and ensures IP address integrity.

Example of DHCP Snooping Configuration:

plaintext Copy code

```
ip dhcp snooping
ip dhcp snooping vlan 2,4,6,10,150
interface FastEthernet0/1
 ip dhcp snooping trust
```

3.5.2 Authentication and Authorization

1. **Authentication:**

   Authentication is the process of verifying the identity of users or devices attempting to access the network. It ensures that only authorized users can access network resources. In our university network
   Ensures that only authorized users can access network resources

2. **Authorization:**

   Authorization determines what an authenticated user is allowed to do on the network. It involves assigning permissions to users based on their roles within the organization, ensuring that users can access only the resources they need to perform their duties.

   Defines user permissions based on roles

3.5.3 Hot Standby Router Protocol (HSRP)

Hot Standby Router Protocol (HSRP) is a Cisco proprietary protocol designed to enhance network reliability and uptime by enabling multiple routers to work together to provide a virtual router as a default gateway. HSRP ensures continuous network availability by facilitating a failover mechanism between routers

Key Benefits of HSRP:

- **Redundancy and Reliability:** Ensures network availability by providing a backup router that can take over if the primary router fails.
- **Minimal Downtime:** Provides seamless transition during router failover, minimizing network disruption.
- **Simplified Configuration:** Allows multiple routers to share the same virtual IP address, simplifying network setup for default gateway configuration

HSRP Components:

1. **Virtual IP Address:** A shared IP address assigned to the HSRP group, used by network hosts as their default gateway.
2. **Priority:** Determines the active router. The router with the highest priority (default is 100) becomes the active router.
3. **Hello Packets:** Sent periodically by routers to indicate their status. The default interval is 3 seconds.

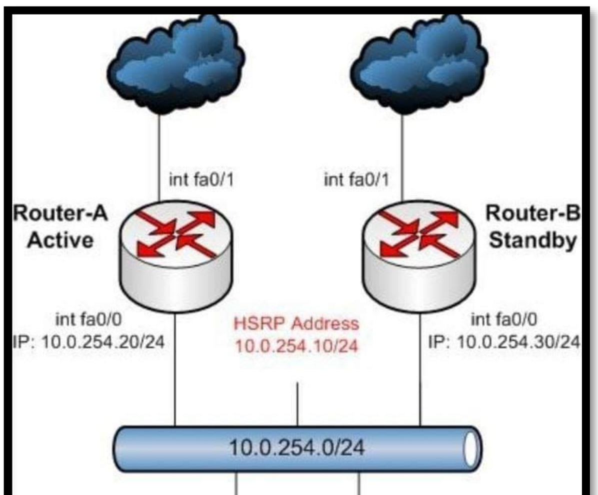
4. **Preemption:** Allows a router with a higher priority to take over as the active router if it becomes available.
5. **Authentication:** Ensures that only authorized routers participate in the HSRP group, preventing unauthorized access.

Types of HSRP States:

- **Initial:** The state when the HSRP process is starting up.
- **Learn:** The router has not yet determined the virtual IP address and is waiting to hear from the active router.
- **Listen:** The router knows the virtual IP address but is neither the active nor the standby router.
- **Speak:** The router sends periodic hello messages and participates in the election process.
- **Standby:** The router is a candidate to become the next active router and monitors the status of the active router.
- **Active:** The router forwards packets sent to the virtual IP address and sends periodic hello messages to inform others of its status.

How HSRP Works:

- **Initial Setup:** Routers in an HSRP group are configured with the same virtual IP address and different priority values.
- **Election Process:** Routers exchange hello packets to elect an active router. The router with the highest priority becomes the active router.
- **Failover:** If the active router fails, the standby router with the next highest priority takes over, ensuring uninterrupted network access.
- **Preemption:** If a higher-priority router becomes available, it can preempt the current active router and take over its role.



➢ Figure 3.15: Diagram showing the HSRP (Hot Standby Router Protocol)

HSRP is used to ensure that the default gateway for network traffic remains available at all times. Multiple routers are configured in an HSRP group, providing redundancy. If the active router fails, the standby router takes over, ensuring that students, faculty, and staff experience no interruption in their network access.

Implementing HSRP in our university's network infrastructure is crucial for achieving high availability and reliability. HSRP allows us to provide a seamless and uninterrupted network experience by ensuring that there is always a backup router ready to take over if the primary router fails. This redundancy is vital for maintaining the continuity of network services and enhancing overall network resilience.

4.0 Introduction

Designing a wireless network that is both robust and efficient is essential for supporting the university's diverse range of activities and users. This chapter focuses on the design and simulation of the wireless network using Packet Tracer, emphasizing Wi-Fi coverage and capacity, particularly with the 802.11ac and 802.11ax standards.

4.1 Wi-Fi Coverage and Capacity (802.11ac, 802.11ax)

802.11ac (Wi-Fi 5):

- Offers high-speed connectivity with data rates up to 1.3 Gbps.
- Features Multi-User MIMO (MU-MIMO) for simultaneous device connections.
- Uses channel bonding (up to 160 MHz) to enhance throughput and reduce congestion.

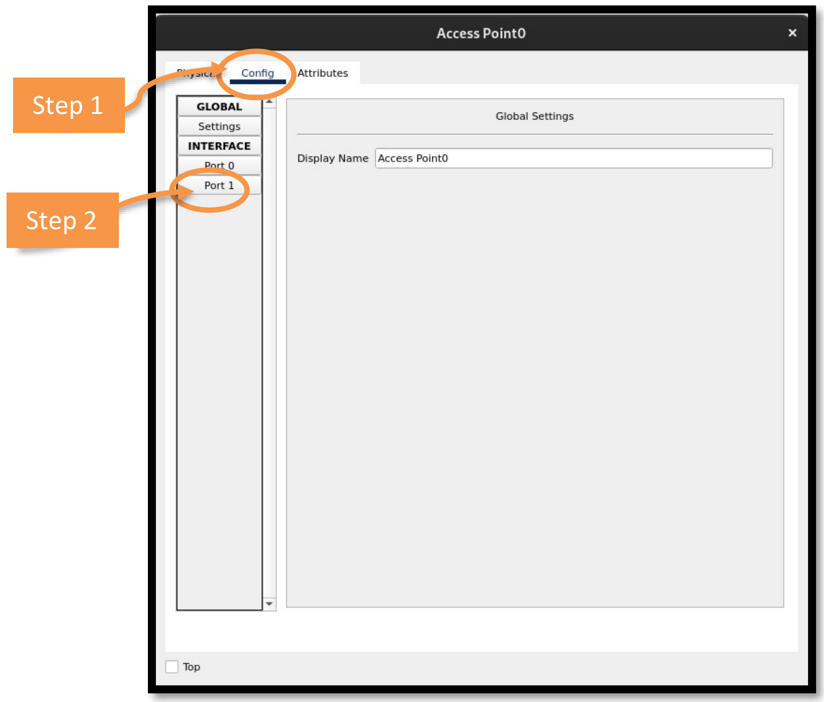
802.11ax (Wi-Fi 6):

- Designed for high-density environments with data rates up to 9.6 Gbps.
- Incorporates OFDMA for efficient simultaneous data transmission to multiple devices.
- Includes Target Wake Time (TWT) to improve battery life by scheduling device wake times.

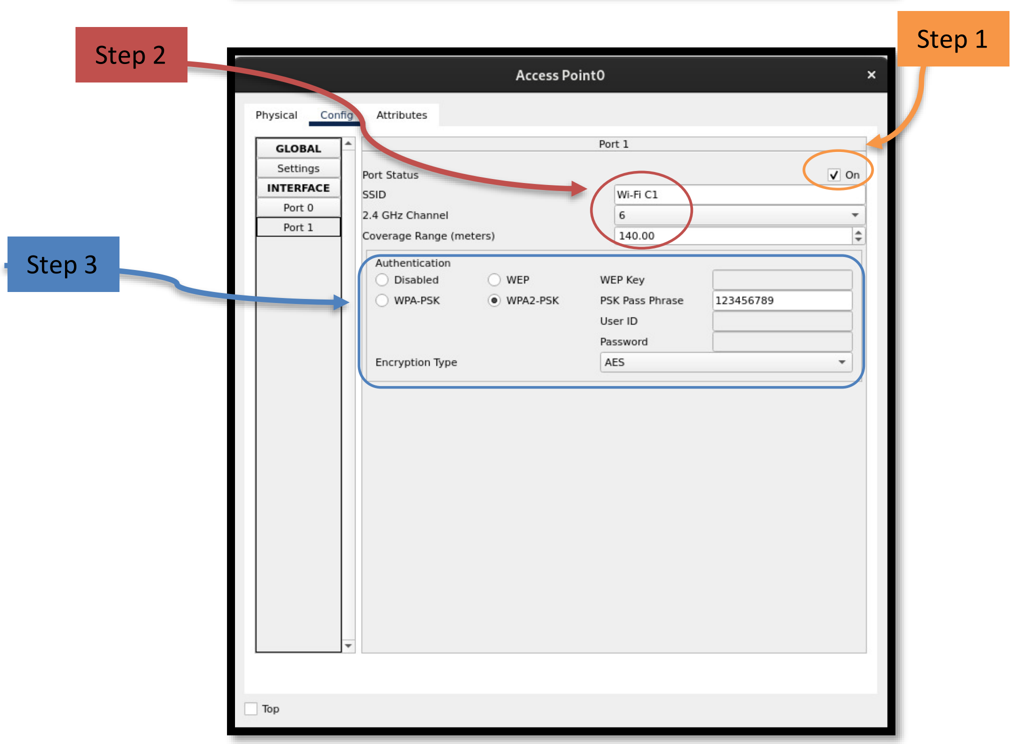
	802.11ac	802.11ax
BANDS	5 GHz	**2.4 GHz** and 5 GHz
CHANNEL BANDWIDTH	20 MHz, 40 MHz, 80 MHz, 80+80 MHz & 160 MHz	20 MHz, 40 MHz, 80 MHz, 80+80 MHz & 160 MHz
FFT SIZES	64, 128, 256, 512	256, 512, 1024, 2048
SUBCARRIER SPACING	312.5 kHz	**78.125 kHz**
OFDM SYMBOL DURATION	3.2 us + 0.8/0.4 us CP	**12.8 us + 0.8/1.6/3.2 us CP**
HIGHEST MODULATION	256-QAM	**1024-QAM**
DATA RATES	433 Mbps (80 MHz, 1 SS) 6933 Mbps (160 MHz, 8 SS)	600.4 Mbps (80 MHz, 1 SS) 9607.8 Mbps (160 MHz, 8 SS)

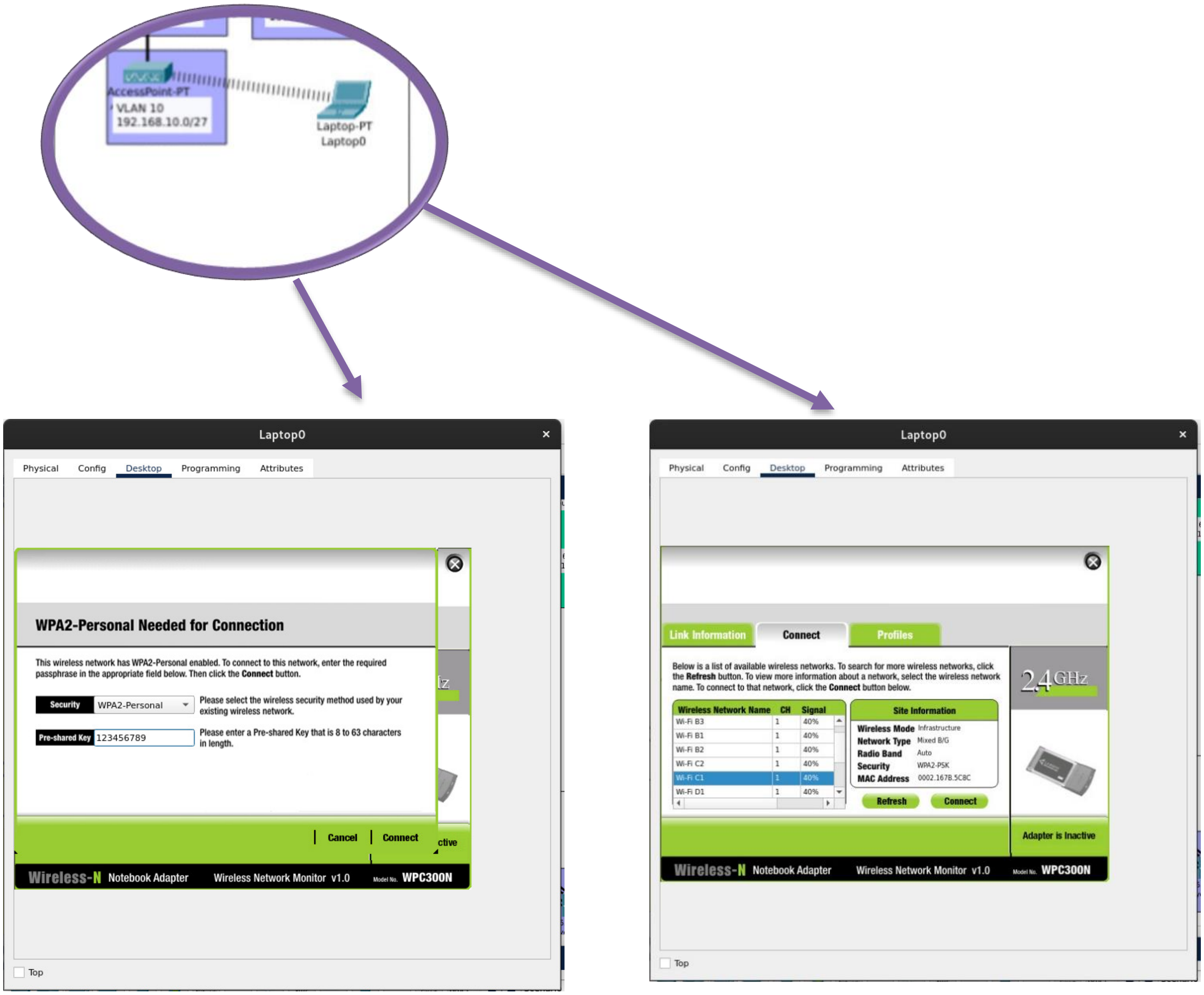
- Table 4.1: the difference between 802.11ac vs. 802.11ax

By leveraging the 802.11ac and 802.11ax standards in Packet Tracer, the wireless network design can ensure comprehensive coverage and sufficient capacity to meet the university's requirements. Effective AP placement, bandwidth management, and interference mitigation are key to providing reliable and high-performing wireless connectivity.



- Figure 4.1: Steps to configure the access point to Wi-Fi





➢ Figure 4.2: Steps to configure Wi-Fi to laptop

5.0 Introduction

Network services and applications are essential components of the university's network infrastructure, enabling communication, collaboration, and research activities. This chapter explores the implementation and benefits of unified communications and the support provided for academic and research applications.

5.1 Unified Communications: VoIP, Video Conferencing

Voice over IP (VoIP)

VoIP enables voice communication over the IP network, offering cost savings, scalability, and integration with other communication tools.

Key Features:

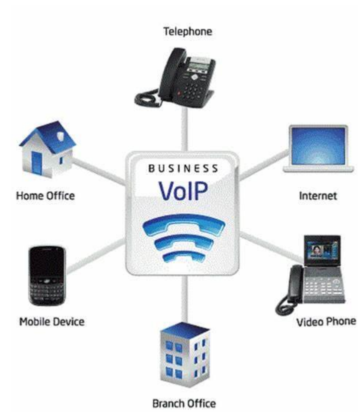
- Cost-effective communication
- Advanced features like voicemail to email and call forwarding
- Integration with email and instant messaging

Video Conferencing

Video conferencing facilitates real-time visual communication, supporting remote teaching, virtual meetings, and collaboration.

Key Features:

- Enhanced remote collaboration
- Accessibility for remote participants
- Reduced travel needs



5.2 Support for Academic and Research Applications

The network supports a variety of academic and research applications, ensuring robust infrastructure for teaching, learning, and innovation.

Learning Management Systems (LMS)

LMS platforms manage coursework and facilitate communication between students and instructors.

Key Features:

- Centralized course materials
- Interactive learning tools
- Accessible from anywhere

Research Applications

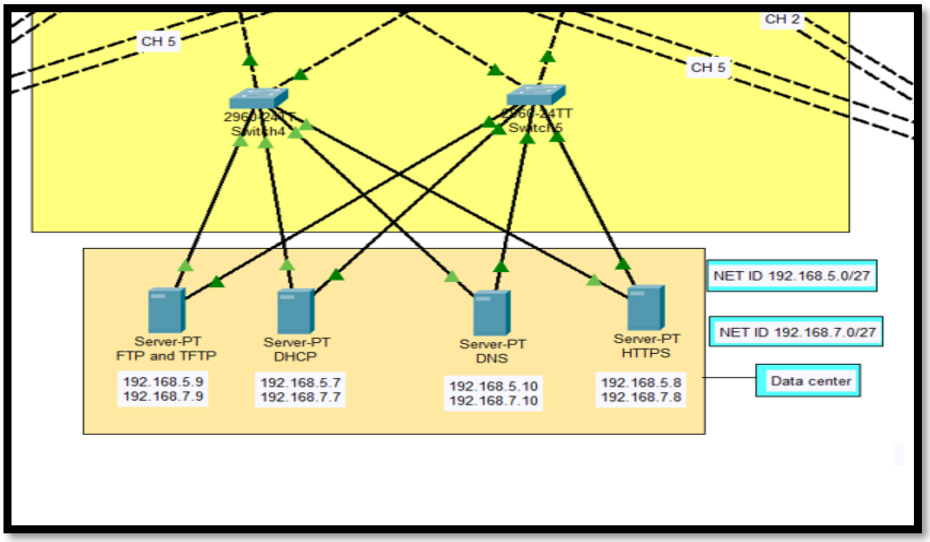
Support for research activities includes data management, collaboration tools, and access to high-performance computing (HPC) resources.

Key Features:

- Efficient data storage and analysis
- Collaborative research tools
- High-speed connectivity for research computing

Unified communications and robust support for academic and research applications ensure the university's network infrastructure meets diverse needs, enhancing communication, collaboration, and innovation.

5.3 Server Implementations



➢ Figure 5.1: Server Implementations

5.3.1 DNS Server

The Domain Name System (DNS) server is critical for resolving human-readable domain names into IP addresses, enabling users to access network resources using friendly names instead of numerical IP addresses.

Functionality:

- Resolves domain names to IP addresses, facilitating easy access to websites and network resources.
- Handles forward and reverse lookups to ensure proper routing and communication between devices.

Benefits:

- **User Convenience:** Simplifies access to university websites and internal resources.
- **Network Management:** Centralizes the management of domain names and IP addresses, making it easier to update and maintain.

- **Efficiency:** Reduces the need for manual IP address entry, minimizing errors and improving user experience.

5.3.2 HTTPS Server

An HTTPS server provides secure communication over the network by encrypting data transmitted between the client and server using SSL/TLS protocols.

Functionality:

- Encrypts data to protect it from interception and tampering.
- Authenticates the server to ensure users are connecting to a legitimate and trusted source.

Benefits:

Data Security: Protects sensitive information such as login credentials and personal data.

Compliance: Ensures adherence to privacy regulations and standards.

Trust: Builds user confidence by providing a secure connection for accessing university services and resources.

5.3.3 FTP Server

File Transfer Protocol (FTP) servers enable the transfer of files between systems on the network, facilitating the uploading, downloading, and management of files.

Functionality:

- Transfers large files efficiently over the network.
- Supports file operations such as uploading, downloading, renaming, and deleting.

Benefits:

- **File Distribution:** Facilitates the distribution of large files such as software updates, research data, and multimedia content.
- **Collaboration:** Enables easy sharing of resources among faculty, students, and staff.
- **Data Management:** Provides a centralized location for managing and archiving important files.

5.3.4 TFTP Server

Trivial File Transfer Protocol (TFTP) servers are used primarily for transferring small configuration files and firmware updates to network devices.

Functionality:

- Transfers configuration files, firmware, and boot files to and from network devices.
- Supports simple file transfer operations without authentication.

Benefits:

- **Device Management:** Streamlines the process of updating firmware and backing up configurations for network devices like routers and switches.
- **Automation:** Facilitates automated network management tasks, reducing the need for manual intervention.
- **Simplicity:** Provides a lightweight and straightforward method for file transfers in controlled environments.

5.3.5 DHCP Server

Dynamic Host Configuration Protocol (DHCP) servers automatically assign IP addresses and other network configuration parameters to devices on the network.

Functionality:

- Assigns dynamic IP addresses to devices, ensuring efficient use of IP address space.
- Provides essential network parameters such as default gateway, subnet mask, and DNS server addresses.

Benefits:

- **Efficiency:** Automates the IP address assignment process, reducing manual configuration efforts and minimizing IP conflicts.
- **Scalability:** Supports a large number of devices, making it easier to scale the network as the university grows.
- **Flexibility:** Allows for easy reconfiguration of network settings without manual intervention on each device.

Chapter 6 Network Management and Monitoring

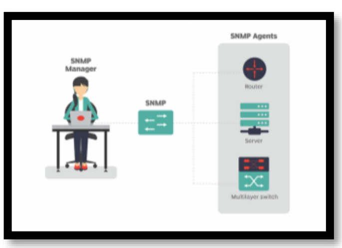
6.0 Introduction

Effective network management and monitoring are essential for maintaining a reliable, secure, and high-performing network infrastructure. This chapter outlines the tools and techniques used for managing and monitoring the university's network, ensuring optimal performance and quality of service.

6.1 Network Management Tools: SNMP, NetFlow, Network Management Systems

Network management tools are crucial for overseeing the network's operation, detecting issues, and ensuring smooth functionality. Key tools include SNMP, NetFlow, and comprehensive Network Management Systems (NMS).

Simple Network Management Protocol (SNMP)



SNMP is a widely used protocol that allows network administrators to monitor and manage network devices. It enables the collection of information about network performance, detection of faults, and configuration of devices through SNMP agents installed on network equipment.

NetFlow

NetFlow is a network protocol developed by Cisco for monitoring and analyzing IP traffic. It provides detailed information about traffic flows, which helps in traffic analysis, security monitoring, and capacity planning.

Network Management Systems (NMS)

NMS platforms, such as SolarWinds and PRTG, offer centralized management and monitoring of network devices. They provide features like device management, real-time performance monitoring, fault detection, and comprehensive reporting and analytics.

6.2 Performance Monitoring and Quality of Service (QoS)

Performance monitoring and QoS are essential for maintaining the quality and reliability of network services, particularly in a university environment where diverse applications and services are used.

Performance Monitoring

Performance monitoring involves tracking and analyzing various network performance metrics to ensure optimal network operation.

Key Metrics:

Bandwidth Utilization: Amount of bandwidth used versus available capacity.

Latency: Time taken for data to travel across the network.

Packet Loss: Percentage of packets lost during transmission.

Jitter: Variability in packet arrival times.

Tools and Techniques:

SNMP and NetFlow: Collect and analyze performance data.

Network Performance Monitoring Software: Tools like SolarWinds, PRTG, and Nagios provide real-time performance insights.

Quality of Service (QoS)

QoS refers to the ability to provide different priority levels for different types of network traffic, ensuring critical applications receive the necessary bandwidth and performance.

QoS Mechanisms:

Traffic Classification: Identifying and categorizing network traffic based on predefined policies.

Traffic Shaping: Regulating the flow of traffic to ensure a smooth and consistent performance.

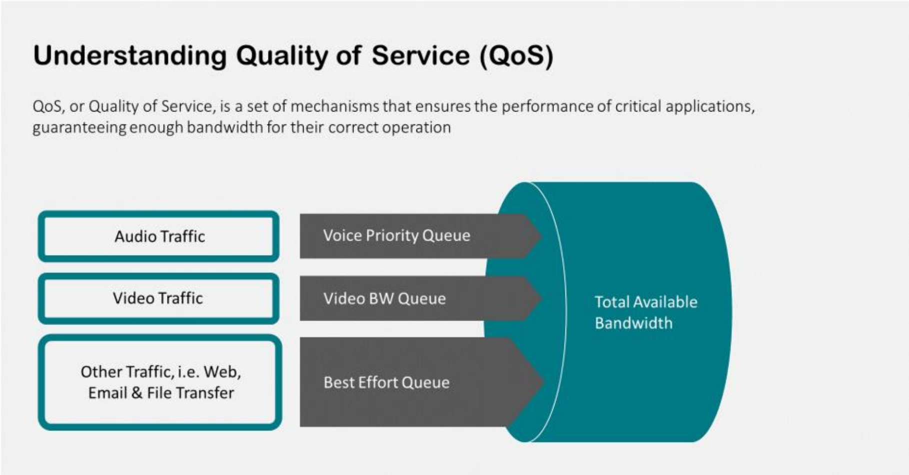
Prioritization: Assigning higher priority to critical traffic, such as VoIP and video conferencing, to ensure low latency and minimal packet loss.

Implementation:

Policy Definition: Establishing QoS policies based on network needs and traffic types.

Configuration: Applying QoS policies to network devices to enforce traffic prioritization and shaping.

Implementing robust network management and monitoring strategies, including the use of SNMP, NetFlow, and advanced Network Management Systems, is crucial for ensuring the university's network remains reliable, secure, and high-performing. Performance monitoring and QoS further enhance the network's ability to meet the diverse and dynamic needs of its users, providing a seamless and efficient network experience.



➢ Figure 6.1: QoS simple Diagram

7.0 Conclusion

In this project, we have meticulously designed and implemented a comprehensive network infrastructure for our university, addressing various critical components such as Ethernet and VLAN configurations, inter-VLAN routing, routing protocols like OSPF, and advanced technologies such as NAT, EtherChannel, and remote access solutions using SSH. The result is a robust, secure, and highly functional network capable of meeting the diverse needs of our academic community.

Key accomplishments include:

VLAN Implementation:
Segregated network traffic into different VLANs for voice, access points, educational purposes, accounts and finance, and administrative offices to enhance performance and security.

Inter-VLAN Routing:
Utilized Layer 3 switches to facilitate efficient and seamless communication between VLANs.

Routing Protocols:
Deployed OSPF within a single are a (Area 0) to ensure dynamic routing and optimal path selection.

Security Measures:
Implemented Access Control Lists (ACLs), port security, DHCP snooping, and authentication and authorization protocols to safeguard network resources.

Addressing and Naming:
Managed IP addresses using /27 subnets and integrated DNS and DHCP for efficient address resolution and assignment

7.1 Integrating All Aspects for a Robust, Secure, and Adaptable Network

The integration of these various components ensures a network that is not only robust and secure but also adaptable to the evolving needs of our university. By combining VLAN segmentation, dynamic routing with OSPF, and stringent security measures, we have created an environment where data integrity, speed, and accessibility are paramount.

Core Integration Points:

Unified VLAN and Routing Strategy: Ensuring smooth communication and data transfer across different departments while maintaining isolation and security where needed.

Comprehensive Security Framework: Implementing layered security protocols across all network levels, from physical devices to data transmission.

Flexible Remote Access: Providing secure, reliable access to network resources for remote users through SSH, enhancing productivity and collaboration

7.2 Ensuring Scalability and Adaptability

Looking towards the future, our network design emphasizes scalability and adaptability to accommodate growth and technological advancements. By planning for expansion and incorporating flexible technologies, we ensure that the network can evolve with the university's needs.

Scalability Strategies:

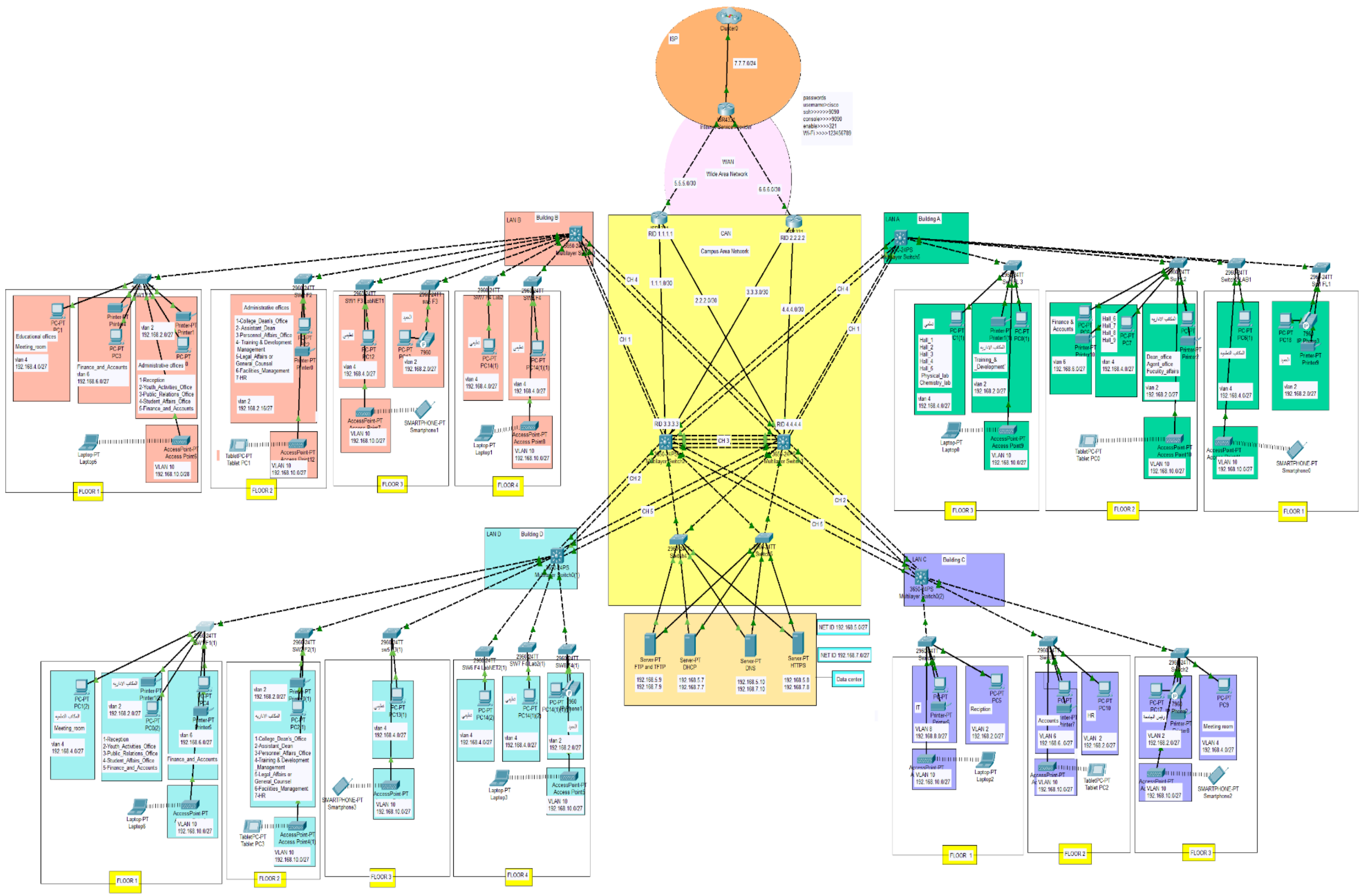
Modular Network Design: Utilizing modular switches and scalable routing protocols to easily add new segments or expand existing ones without significant reconfiguration.

Future-Proof Technologies: Implementing technologies like EtherChannel and PAT to handle increasing data loads and diverse application requirements.

Continuous Monitoring and Upgrades: Establishing a framework for regular network performance monitoring, security audits, and timely upgrades to keep the infrastructure up-to-date with emerging threats and technological trends.

By integrating all these aspects, we have laid the groundwork for a network that not only meets current demands but is also prepared for future challenges. Our approach ensures that the university network remains a robust, secure, and adaptable backbone for all academic, administrative, and research activities.

Network Diagram:



Appendices

Appendix A: Network Design Diagrams

This appendix includes detailed diagrams of the network design, illustrating VLAN configurations, inter-VLAN routing, OSPF area setup, NAT implementation, and remote access configurations.

Appendix B: Configuration Examples

Here, we provide example configurations for various network devices, including switches and routers, to help replicate the network design.

Appendix C: Security Policies

This appendix contains the security policies and procedures implemented in the network, including ACLs, port security, and DHCP snooping configurations.

Appendix D: IP Address Allocation Table

A detailed table showing the IP address allocation for different subnets and VLANs, including the /27 subnets used in the network.

Glossary of Terms

ACL (Access Control List):

A set of rules used to control network traffic and reduce network attacks.

DHCP (Dynamic Host Configuration Protocol):

A network management protocol used to automate the process of configuring devices on IP networks.

DNS (Domain Name System):

A hierarchical system for naming resources on the Internet.

EtherChannel:

A port link aggregation technology used to combine multiple physical links into a single logical link for increased bandwidth and redundancy.

HSRP (Hot Standby Router Protocol):

A Cisco protocol used to provide high network availability by routing IP traffic without any dependence on the availability of any single router.

NAT (Network Address Translation):

A method of remapping one IP address space into another.

OSPF (Open Shortest Path First):

A routing protocol for Internet Protocol (IP) networks that uses a link-state routing algorithm.

PAT (Port Address Translation):

A type of NAT that allows multiple devices on a local network to be mapped to a single public IP address but with a different port number for each session.

SSH (Secure Shell):

A protocol for securely accessing and managing devices over an unsecured network.

STP (Spanning Tree Protocol):

A network protocol that ensures a loop-free topology for Ethernet networks. It prevents network loops by dynamically blocking redundant paths in the network topology

VLAN (Virtual Local Area Network):

A method to create multiple distinct broadcast domains that are mutually isolated.

Sources and Further Reading Recommendations

Books:

"Computer Networking: A Top-Down Approach" by James F. Kurose and Keith W. Ross

"TCP/IP Illustrated, Volume 1: The Protocols" by W. Richard Stevens

"Routing TCP/IP, Volume 1" by Jeff Doyle and Jennifer DeHaven Carroll

"Network Warrior" by Gary A. Donahue

Articles and Papers:

"Designing Large-Scale LANs" by Cisco Systems

"The Implementation and Benefits of VLANs in Enterprise Networks" by IEEE Communications Society

"Security Considerations in VLAN and Inter-VLAN Routing" by Network World

Websites:

Cisco Networking Academy

https://www.netacad.com/

Network World

https://www.networkworld.com/

IEEE Communications Society

https://www.comsoc.org/